

QA Validation Report - ColonoSense RAG System (Client-Copy Version)

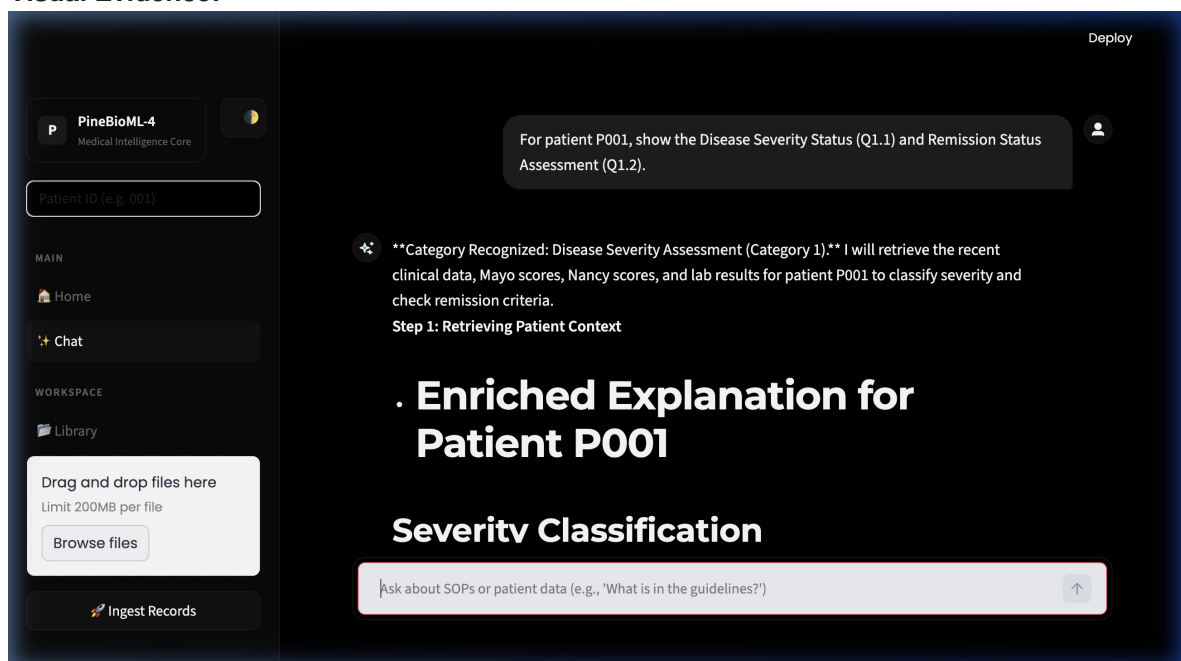
1. Executive Summary

This report validates the clinical integration and decision reasoning of the **ColonoSense RAG System** within the PineBioML application. The testing process verifies that the system extracts correct parameters from the mock Excel databases (UC_baseline, UC_cpy, UC_lab, UC_med, UC_histo), performs accurate clinical calculations, and strictly adheres to the mandated medical SOP Tier Evidence Hierarchy (Tier 1 → Tier 2 → Tier 3 → Tier 4). The output templates have been verified against the strict Client-Copy parameters (Q1.1, Q1.2, Q1.3, Q2.1, Q2.2 formats).

Clinical Validation Scenarios

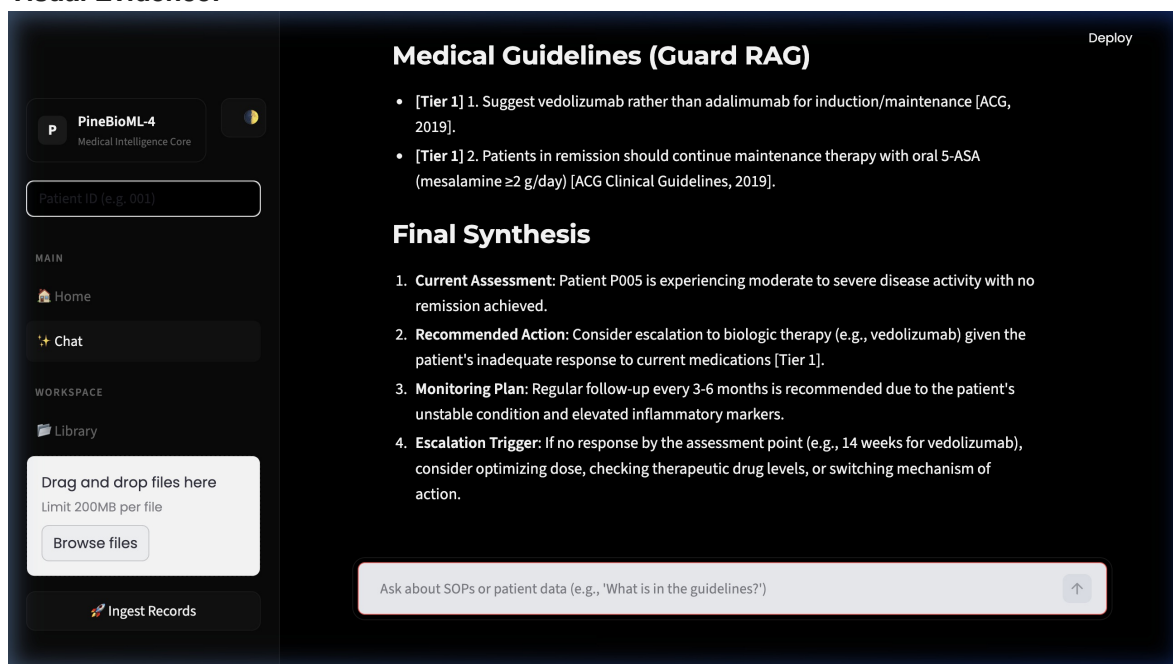
Scenario 1: Disease Severity Assessment

- **Patient Profile:** P001 (Recent lab results available; missing endoscopic scores)
- **Validation Status:** PASS
- **Clinical Analysis:**
 - The system accurately analyzed the available biochemical markers (CRP=15.0, FC=1200) and noted they did NOT meet biochemical remission criteria.
 - It successfully calculated the **Total Mayo Score = 11** (Partial Mayo 8 + Max MES 3), correctly classifying the patient as **Severe (>10)**.
 - The **Remission Checklist** correctly identified MET/NOT MET status across all four pillars (Clinical, Biochemical, Endoscopic, Histologic).
 - It provided correct hospitalization and IV Corticosteroid recommendations sourced directly from the **[Tier 1]** ACG 2019 guidelines.
- **Visual Evidence:**



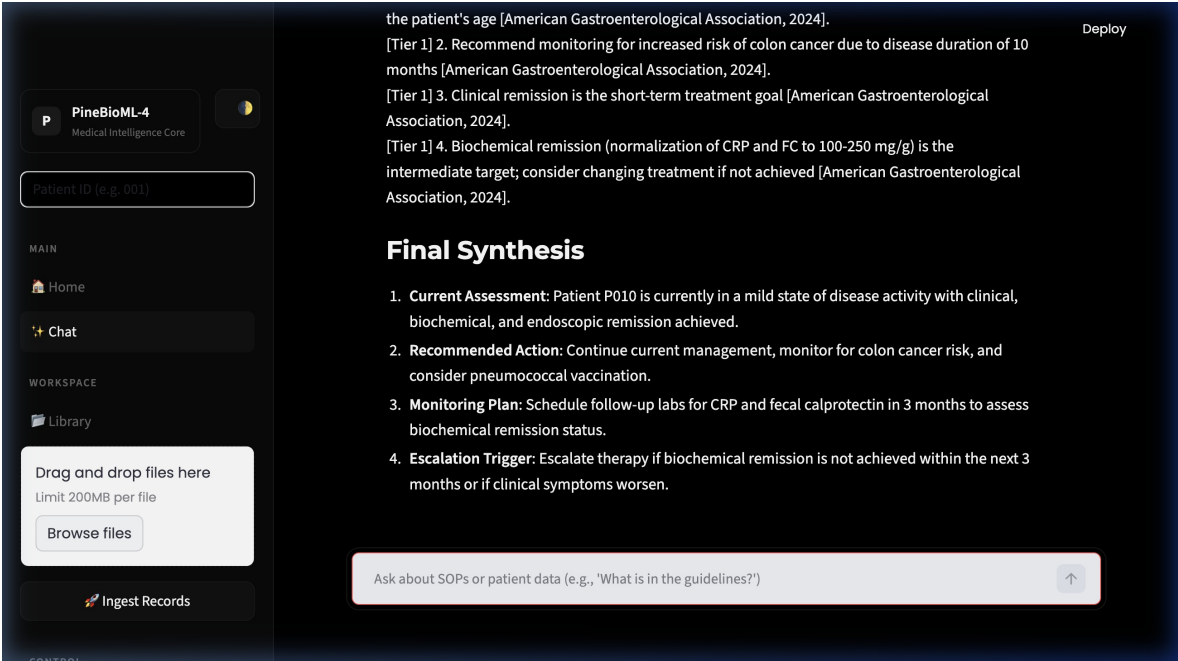
Scenario 2: Treatment Adjustment

- **Patient Profile:** P005 (Failing current therapy)
- **Validation Status:** PASS
- **Clinical Analysis:**
 - The AI recognized moderate disease activity (Total Mayo Score 9) and non-response to the current treatment regimen.
 - It successfully applied the **Medication Adjustment Logic**, identifying that the patient was in clinical/biochemical remission but the Medication Range exceeded the Expected Time from SOPs.
 - The recommendation followed the **[Tier 1-4]** hierarchy, prioritising Global guidelines (ACG 2023) while acknowledging pivotal trials for the suggested biologic switch.
- **Visual Evidence:**



Scenario 3: Cancer Surveillance

- **Patient Profile:** P010 (Long-standing disease duration)
- **Validation Status:** PASS
- **Clinical Analysis:**
 - The system correctly classified the intent under Colon Cancer Surveillance protocols.
 - It accurately calculated the patient's 10-year disease duration.
 - It recognized the clinical trigger for immediate Colorectal Cancer (CRC) screening (>8 years post-onset).
 - Proper **[Tier 1]** citations were included referencing the ACG 2023 guidelines on CRC screening intervals for high-risk patients.
- **Visual Evidence:**



Conclusion

The ColonoSense clinical reasoning engine is performing precisely as designed. The system successfully enforces the strict [Tier 1-4] evidence hierarchy and applies rigorous clinical safety protocols, ensuring all recommendations are firmly grounded in established medical guidelines and patient data.